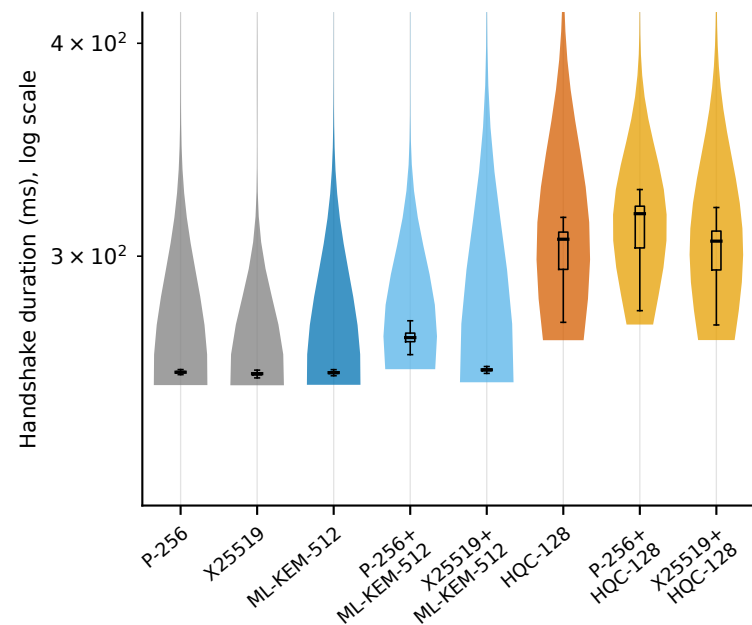
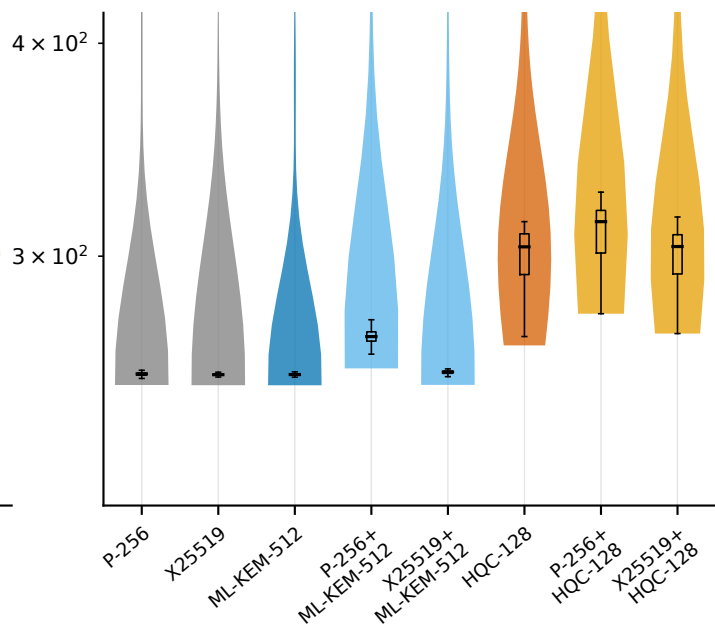


(a)

ed25519 (classical)

**(b)**

mldsa44 (post-quantum)

**(c)**

Classical KEX
 ML-KEM (pure)
 ML-KEM (hybrid)
 HQC (pure)
 HQC (hybrid)

KEM	ed25519 Mean (ms)	ed25519 CV (%)	ed25519 Max (ms)	mldsa44 Mean (ms)	mldsa44 CV (%)	mldsa44 Max (ms)
P-256	275.5	43.9	1.31k	278.3	46.6	1.44k
X25519	266.1	33.3	1.32k	288.3	55.0	1.31k
ML-KEM-512	273.9	43.9	1.30k	276.5	39.8	1.31k
P-256+ML-KEM-512	282.5	36.5	1.42k	302.1	57.7	2.34k
X25519+ML-KEM-512	284.8	60.6	2.31k	290.8	53.9	1.44k
HQC-128	332.3	46.5	1.79k	335.8	47.1	1.35k
P-256+HQC-128	331.8	31.8	1.36k	345.1	55.9	3.39k
X25519+HQC-128	329.1	39.7	1.35k	332.5	48.6	2.30k